MILITARY SPECIFICATION SHEET

CONNECTOR, COMPONENT PARTS, ELECTRICAL BACKPLANE, PRINTED WIRING, LOW INSERTION FORCE (LIF), COMPLIANT CONTACT

This specification sheet is approved for use by the Naval Sea Systems Command, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the compliant contact described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-28859.

AMSC N/A FSC 5935
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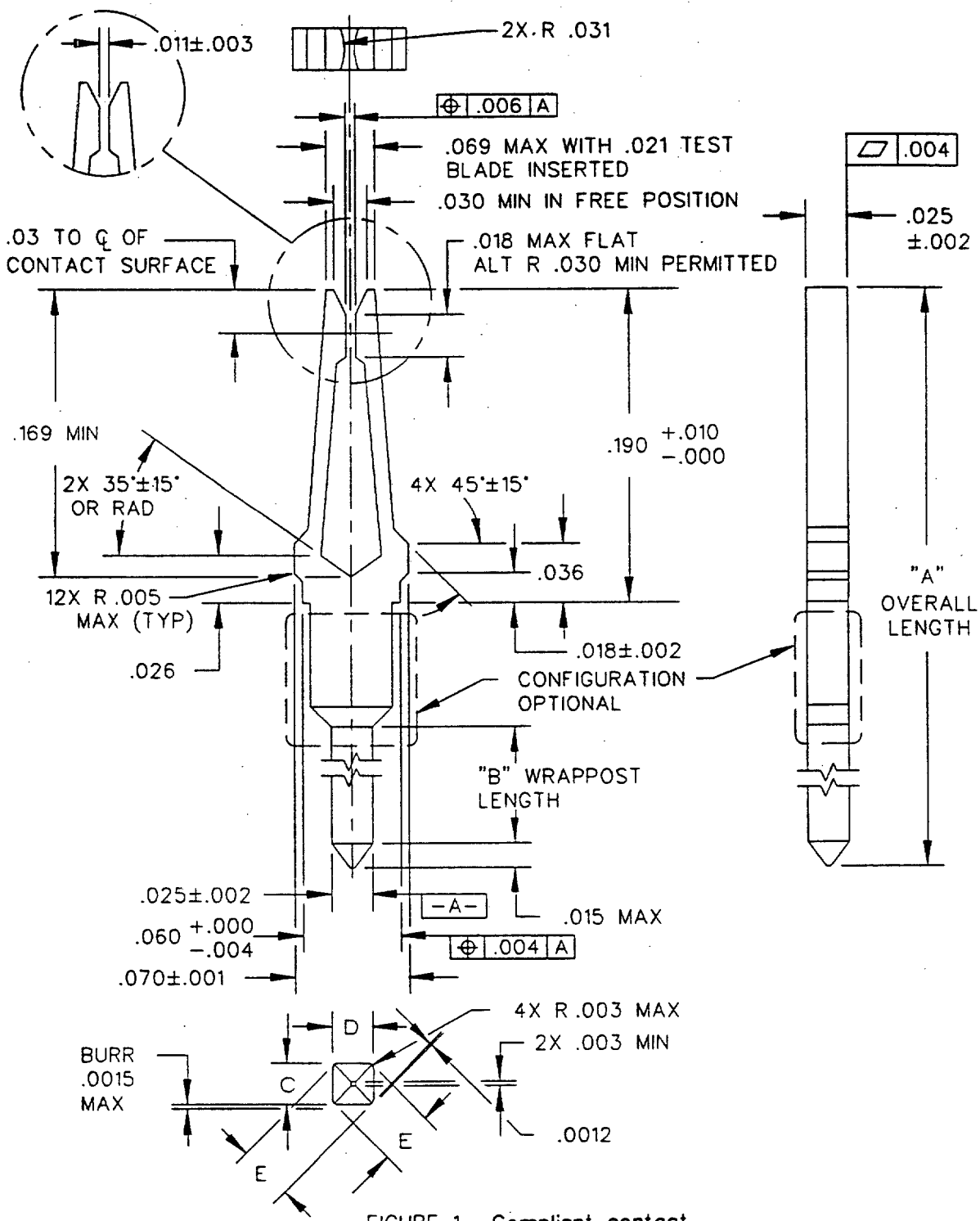


FIGURE 1. Compliant contact.

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Wrappost Geometry

C	D	E	Parallelism	Straightness
0.025(NOM)	0.025(NOM)	0.0355MAX	0.002	0.005in/in
0.023MIN	0.023MIN	0.0325MIN		

NOTES:

1. Part to be free of burrs and sharp edges except as noted.
2. Dimensions apply after plating.
3. Dimensions to top of tines shall be measured to highest tine.
4. Unless otherwise specified, tolerances are ±0.01 inch (0.3 mm) on two place decimals; ±0.005 inch (0.13 mm) on three place decimals; and ±0.0001 inch (0.003 mm) on four place decimals.

FIGURE 1. Compliant contact - Continued.

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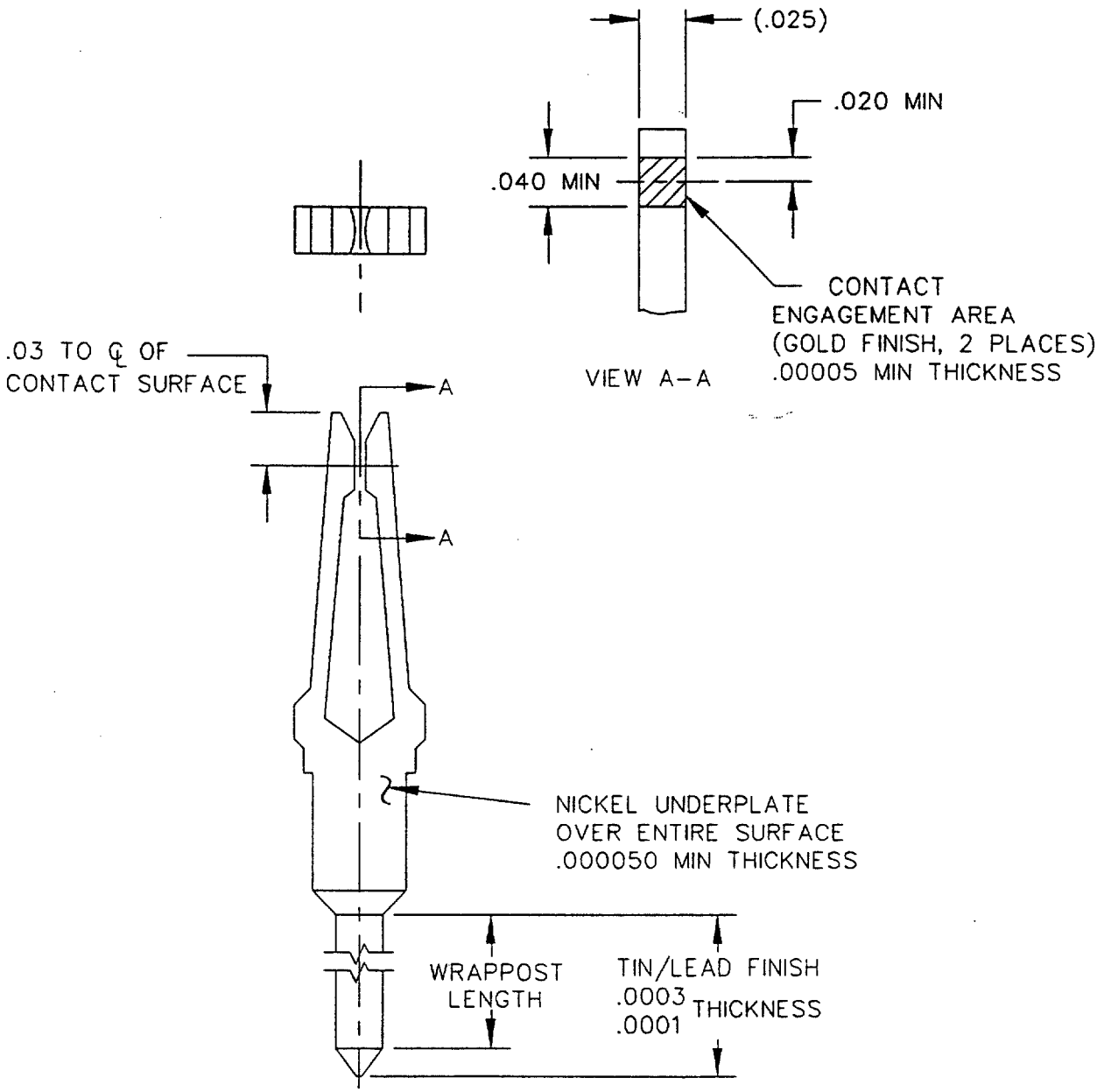


FIGURE 2. Localized finish.

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INCHES	MM
0.00005	0.00127
0.0001	0.0025
0.0003	0.0076
0.010	0.254
0.020	0.508
0.025	0.635
0.03	0.76
0.040	1.016

NOTES:

1. Dimensions are in inches. Metric equivalents are given for information only.

2. Dimensions in this figure are for localizing areas of localized finish only.

3. Unless otherwise specified, tolerances are ±0.01 inch (0.3 mm) on two place decimals; ±0.005 inch (0.13 mm) on three place decimals; and ±0.0001 inch (0.003 mm) on four place decimals.

FIGURE 2. Localized finish - Continued.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Material: Cu-Be alloy, ASTM B194, UNS No. C17200, with a minimum Knoop microhardness of 310.

Housing: In accordance with MIL-C-28859/2.

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Finish: Gold plate, MIL-G-45204, type II, grade C, class 1, 50 microinches minimum over nickel plating, QQ-N-290, class 1, 50 microinches minimum.

Localized finish: Contacts supplied with a localized finish shall meet the requirements of MIL-C-28859 and this document. Localized finish contacts shall conform to figures 1 and 2. Nickel plating shall meet QQ-N-290, class 1, 50 microinches minimum over the entire contact. Gold in accordance with MIL-G-45204, type II, grade C, class 1, 50 microinches on the area specified on figure 2. Tin-lead finish in accordance with MIL-P-81728 (tin composition 50 to 95 percent). Tin-lead thickness shall be 0.0001 to 0.0003 inch (0.0025 to 0.0076 mm) thick.

Current rating: 3 amps.

Military part number: M28859/- (dash number from table I). Contacts with localized finish shall have an S following the dash number. For example M28859/5-02S.

Insertion force: Average insertion force shall be 2.25 ounces maximum with random readings at 4.00 ounces maximum.

Separation: 4.0 ounces maximum.

Normal force: The average normal force on each tine shall be greater than 3.53 ounces with random readings of not less than 3.00 ounces.

QUALIFICATIONS:

Localized contact finish: The qualification of the contact supplied with the standard finish does not extend qualification to the contact supplied with the localized finish. To be qualified to supply the localized finish in lieu of the standard contact finish, manufacturers shall subject the contact supplied with localized finish to the same qualification procedures as the standard contact finish. Manufacturers that have qualified the localized finish contact will be designated on the Qualified Products List (QPL).

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TABLE I. Dash number and dimensions.

Dash No.	Overall length A ±0.010	Wrappost length B (min)
01	0.375 Inch	0 Inch
02	0.550	0.175
03	0.625	0.250
04	0.725	0.350
05	0.875	0.500
06	1.000	0.625
07	1.125	0.750

Custodian:
Navy - SH

User activity:
Navy - AS

Preparing activity:
Navy - SH

Agent: NW

(Project 5935-N365)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-28859/5(SH)	Connector, Component Parts, Electrical Backplane, Printed Wiring, Low Insertion Force (LIF), Compliant Contact

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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